```
import numpy as np
import pandas as pd
import matplotlib as mpl
import matplotlib
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
```

2.2 读入数据并进行初步 EDA 操作

```
train = pd.read_csv('/home/aistudio/data/data101120/categorical_train.csv')
test = pd.read_csv('/home/aistudio/data/data101120/categorical_test.csv')
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'pd' is not defined
```

```
import pandas as pd
train = pd.read_csv('/home/aistudio/data/data101120/categorical_train.csv')
test = pd.read_csv('/home/aistudio/data/data101120/categorical_test.csv')
```

```
train.shape, test.shape
```

```
((300000, 40), (200000, 39))
```

2.3 查看数据类型及缺失情况

```
train.info()
test.info()
```

```
<class'pandas.core.frame.DataFrame'>
RangeIndex: 300000 entries, 0 to 299999
Data columns (total 40):
id          300000 non-null int64
feature_1   300000 non-null int64
feature_2   300000 non-null int64
feature_3   300000 non-null int64
feature_4   300000 non-null int64
feature_5   300000 non-null int64
feature_6   300000 non-null int64
feature_7   300000 non-null int64
feature_8   300000 non-null int64
feature_9   300000 non-null int64
feature_10  300000 non-null int64
feature_11  300000 non-null int64
feature_12  300000 non-null int64
feature_13  300000 non-null int64
feature_14  300000 non-null int64
feature_15  300000 non-null int64
feature_16  300000 non-null int64
feature_17  300000 non-null int64
feature_18  300000 non-null int64
feature_19  300000 non-null int64
feature_20  300000 non-null int64
feature_21  300000 non-null int64
feature_22  300000 non-null int64
feature_23  300000 non-null int64
feature_24  300000 non-null int64
feature_25  300000 non-null int64
feature_26  300000 non-null int64
feature_27  300000 non-null int64
feature_28  300000 non-null int64
feature_29  300000 non-null int64
feature_30  300000 non-null int64
feature_31  300000 non-null int64
feature_32  300000 non-null int64
feature_33  300000 non-null int64
feature_34  300000 non-null int64
feature_35  300000 non-null int64
feature_36  300000 non-null int64
feature_37  300000 non-null int64
feature_38  300000 non-null int64
feature_39  300000 non-null int64
dtypes: int64(40)
memory usage: 91.5 MB
<class'pandas.core.frame.DataFrame'>
RangeIndex: 200000 entries, 0 to 199999
Data columns (total 39):
id          200000 non-null int64
feature_1   200000 non-null int64
feature_2   200000 non-null int64
feature_3   200000 non-null int64
feature_4   200000 non-null int64
feature_5   200000 non-null int64
feature_6   200000 non-null int64
feature_7   200000 non-null int64
feature_8   200000 non-null int64
feature_9   200000 non-null int64
feature_10  200000 non-null int64
feature_11  200000 non-null int64
feature_12  200000 non-null int64
feature_13  200000 non-null int64
feature_14  200000 non-null int64
feature_15  200000 non-null int64
feature_16  200000 non-null int64
feature_17  200000 non-null int64
feature_18  200000 non-null int64
feature_19  200000 non-null int64
feature_20  200000 non-null int64
feature_21  200000 non-null int64
feature_22  200000 non-null int64
feature_23  200000 non-null int64
feature_24  200000 non-null int64
feature_25  200000 non-null int64
feature_26  200000 non-null int64
feature_27  200000 non-null int64
feature_28  200000 non-null int64
feature_29  200000 non-null int64
feature_30  200000 non-null int64
feature_31  200000 non-null int64
feature_32  200000 non-null int64
feature_33  200000 non-null int64
feature_34  200000 non-null int64
feature_35  200000 non-null int64
feature_36  200000 non-null int64
feature_37  200000 non-null int64
feature_38  200000 non-null int64
feature_39  200000 non-null int64
dtypes: int64(39)
memory usage: 59.6 MB
```

2.4 查看数据分布

```
train.describe
```

```
<bound method NDFrame.describe of          id  feature_1  feature_2  feature_3  feature_4  feature_5  feature_6  feature_7  feature_8  feature_9  ...  feature_30  feature_31  feature_32  feature_33  feature_34  feature_35  feature_36  feature_37  feature_38  feature_39
0         0          1          1          1          1          1          1          1          1          1  ...           1           1           1           1           1           1           1           1           1           1
1         1          1          1          1          1          1          1          1          1          1  ...           1           1           1           1           1           1           1           1           1           1
2         2          1          1          1          1          1          1          1          1          1  ...           1           1           1           1           1           1           1           1           1           1
3         3          1          1          1          1          1          1          1          1          1  ...           1           1           1           1           1           1           1           1           1           1
4         4          1          1          1          1          1          1          1          1          1  ...           1           1           1           1           1           1           1           1           1           1
...     ...        ...        ...        ...        ...        ...        ...        ...        ...        ...  ...         ...         ...         ...         ...         ...         ...         ...         ...         ...
299995  299995        1          1          1          1          1          1          1          1          1  ...           1           1           1           1           1           1           1           1           1           1
299996  299996        1          1          1          1          1          1          1          1          1  ...           1           1           1           1           1           1           1           1           1           1
299997  299997        1          1          1          1          1          1          1          1          1  ...           1           1           1           1           1           1           1           1           1           1
299998  299998        1          1          1          1          1          1          1          1          1  ...           1           1           1           1           1           1           1           1           1           1
299999  299999        1          1          1          1          1          1          1          1          1  ...           1           1           1           1           1           1           1           1           1           1

[300000 rows × 40 columns]>
```

2.5 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.6 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.7 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.8 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.9 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.10 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.11 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.12 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.13 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.14 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.15 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.16 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.17 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.18 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.19 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.20 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.21 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.22 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.23 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.24 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.25 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.26 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.27 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.28 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.29 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.30 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.31 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.32 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.33 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.34 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.35 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.36 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.37 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.38 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.39 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.40 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.41 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.42 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.43 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.44 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.45 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.46 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.47 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.48 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.49 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.50 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.51 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.52 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.53 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.54 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.55 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.56 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.57 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.58 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.59 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.60 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.61 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.62 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.63 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.64 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.65 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.66 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.67 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.68 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.69 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.70 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.71 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.72 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.73 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.74 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.75 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.76 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.77 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.78 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.79 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.80 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.81 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.82 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.83 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.84 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.85 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.86 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.87 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.88 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.89 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.90 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.91 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.92 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.93 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.94 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.95 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.96 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.97 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.98 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.99 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.100 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.101 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.102 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.103 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.104 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.105 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.106 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.107 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.108 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.109 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.110 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.111 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.112 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.113 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.114 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.115 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.116 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.117 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.118 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.119 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.120 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.121 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.122 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.123 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.124 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.125 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.126 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.127 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.128 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.129 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.130 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.131 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.132 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.133 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.134 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.135 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.136 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.137 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.138 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.139 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.140 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.141 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.142 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.143 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.144 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.145 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.146 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.147 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.148 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.149 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.150 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.151 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.152 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.153 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.154 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.155 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.156 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.157 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.158 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.159 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.160 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.161 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.162 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.163 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.164 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.165 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.166 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.167 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.168 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.169 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.170 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.171 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.172 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.173 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.174 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.175 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.176 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.177 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.178 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.179 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.180 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.181 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.182 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.183 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.184 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.185 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.186 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.187 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.188 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.189 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.190 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.191 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.192 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.193 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.194 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.195 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'train' is not defined
```

2.196 查看数据分布

```
train.describe()
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name '
```